

STA304 Practice Final Examination

Time: 3 Hours
Total Marks: 100

Instructions:

- Show all work.
 - Clearly state any assumptions.
 - Unless otherwise stated, assume sampling is without replacement.
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Question 1 (10 marks): Concepts and Definitions

Answer each part in one or two sentences.

- a) Define a sampling frame.

- b) Explain the difference between a target population and a sampled population.

- c) Define MCAR.

- d) Define MAR.

- e) Define NMAR.

- f) Explain why a low response rate does not necessarily imply high nonresponse bias.

- g) State one advantage of stratified sampling.

- h) State one advantage of cluster sampling.

i) Explain what a design weight represents.

j) Define a domain estimator.

Question 2 (10 marks): Simple Random Sampling

A simple random sample of $n = 100$ households is selected from a population of $N = 2000$ households. Let y_i denote monthly electricity usage, measured in kWh.

- a) What is the probability that a particular household is selected?
- b) Write an estimator for the population mean electricity usage.
- c) Suppose the sample mean is $\bar{y}_S = 680$. Estimate the population total electricity usage.
- d) Compute the finite population correction.
- e) A student claims that increasing the population size while holding n fixed always decreases the variance of $\bar{y}_S$.
Is the student correct? Explain.

Question 3 (8 marks): SRS Theory

Let

$$I_i = \begin{cases} 1, & \text{if unit } i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

a) Find $E(I_i)$.

b) Interpret this quantity.

c) Show that

$$\hat{t}_y = \frac{N}{n} \sum_{i \in S} y_i$$

is unbiased for the population total t_y .

Question 4 (14 marks): Stratified Sampling

A population is divided into three strata.

Stratum	N_h	S_h
1	1000	8
2	2000	20
3	3000	30

A sample of size $n = 300$ will be selected.

a) Find the proportional allocation.

b) Find the Neyman allocation.

c) Which stratum receives the largest increase under Neyman allocation? Explain why.

d) Explain why stratification may improve precision in this setting.

e) Describe a situation where proportional allocation might be preferred over Neyman allocation.

Question 5 (10 marks): Survey Design and Selection Bias

For each scenario, identify the primary source of selection bias and briefly explain your answer.

Possible answers:

- Undercoverage
 - Convenience sampling
 - Self-selection
 - Nonresponse
 - None of the above
- a) A university conducts a survey by placing a link on its homepage and inviting students to participate.
- b) A telephone survey uses a directory containing only landline numbers.
- c) A survey of restaurant satisfaction is conducted only among customers who leave online reviews.
- d) A mail survey achieves a response rate of 35%.
- e) A researcher interviews the first 50 people leaving a shopping mall.

Question 6 (12 marks): Cluster Sampling

A city contains 500 apartment buildings. A sample of 20 buildings is selected. Within each sampled building, 15 residents are sampled.

Let $\bar{y}_i$ denote the sample mean satisfaction score within building i .

- a) Identify the PSUs and SSUs.

- b) Explain why this design may be more convenient than an SRS of residents.

- c) Suppose the following building means are observed:

70, 75, 82, 79, 68, 73, 77, 71, 80, 74,
72, 83, 76, 69, 78, 74, 81, 70, 75, 79.

Estimate the overall population mean.

- d) Explain why cluster sampling may be less efficient than an SRS of the same number of residents.

- e) If the ICC were 0, what would this imply about the efficiency of cluster sampling?

Question 7 (12 marks): Unequal Probability Sampling

A wildlife agency wishes to estimate the total number of nesting birds in four wetlands.

Wetland	Area (hectares)	Bird Count
A	100	120
B	300	350
C	600	700
D	1000	1150

One wetland is selected using PPS sampling with replacement.

- a) Compute the selection probability of Wetland C.
- b) Suppose Wetland C is selected. Compute the estimate of the total.
- c) Suppose Wetland A is selected instead. Compute the estimate of the total.
- d) Explain why PPS sampling may be preferable to equal probability sampling.

e) What characteristic of the size measure makes PPS particularly effective?

Question 8 (6 marks): Ratio Estimation

A simple random sample of 50 farms is selected.

The sample totals are

$$t_x = 6000, \quad t_y = 8400.$$

The known population total acreage is

$$X = 72,000.$$

- a) Compute the estimated ratio.
- b) Estimate the population total crop yield.
- c) Suppose the sample underestimates the population mean acreage and x and y are positively correlated.
Would the ratio estimator likely be larger or smaller than the expansion estimator? Explain.

Question 9 (10 marks): Nonresponse and Missing Data

A survey of exercise habits obtains a 75% response rate. Administrative records provide age group information for both respondents and nonrespondents.

Age Group	Sampled	Respondents
18–34	80	72
35–54	70	56
55+	50	22

- Compute the response rate within each age group.
- Which age group appears most problematic?
- Explain why this pattern could create nonresponse bias.
- Describe one strategy for adjusting for this nonresponse.
- Would weighting adjustment require information about nonrespondents? Explain.

Question 10 (8 marks): Post-Stratification and Domain Estimation

A survey estimates mean weekly exercise hours.

The respondent sample is

Age Group	Sample Mean
18–34	5.0
35–54	4.0
55+	3.0

Known population proportions are

Age Group	Population Proportion
18–34	0.25
35–54	0.45
55+	0.30

- a) Compute the post-stratified estimate of the population mean.
- b) Explain why post-stratification may reduce bias.
- c) What additional information is required to perform post-stratification?
- d) Explain the difference between a domain estimate and a post-stratified estimate.

Question 11 (10 marks): Survey Design Case Study

A public health agency wishes to estimate the proportion of adults who received a flu vaccine in the past year.

The survey will be conducted using in-person interviews and has a limited budget.

- a) Propose a sampling design.
- b) Identify the target population.
- c) Identify a suitable sampling frame.
- d) Describe one potential sources of bias.

- e) Suggest one method for reducing this source of bias.

End of Examination